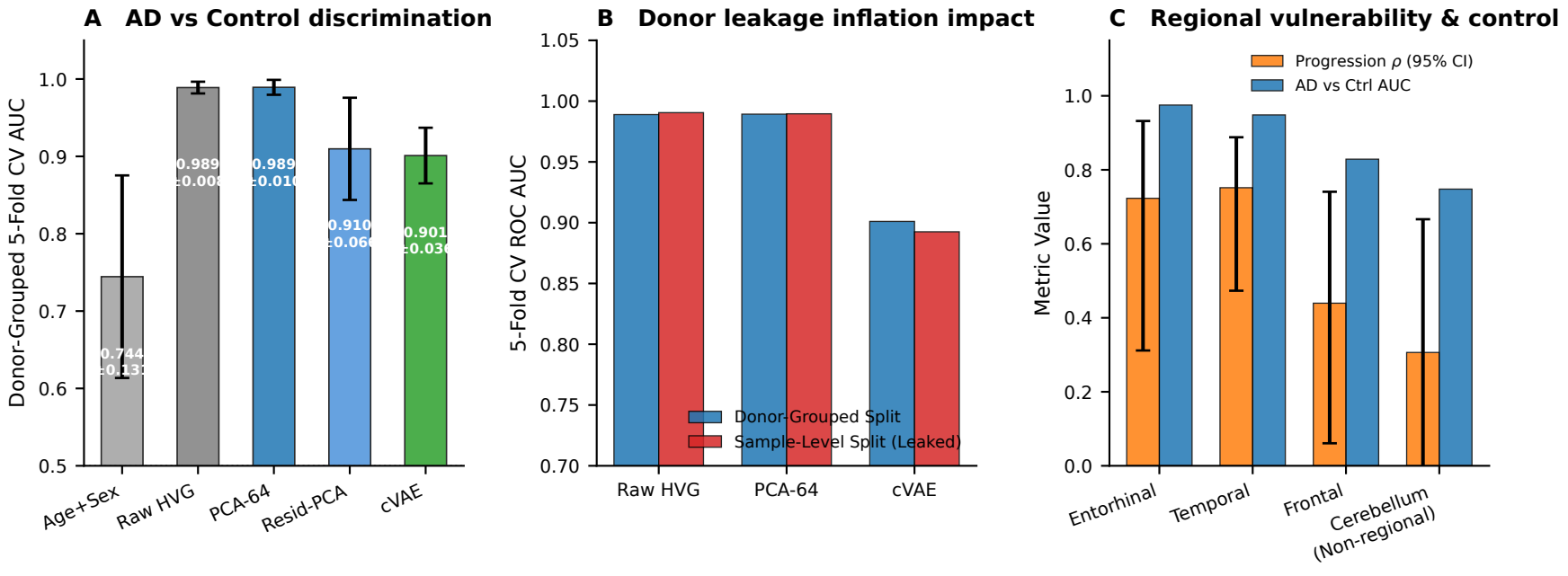
A AD vs Control discrimination
Donor-Grouped 5-Fold CV AUC
1.0
0.9
0.8
0.7
0.6
0.5
0.744
±0.13
0.989
±0.00
0.989
±0.01
0.910
±0.06
0.901
±0.03
Age+Sex
Raw HVG
PCA-64
Resid-PCA
cVAE
B Donor leakage inflation impact
5-Fold CV ROC AUC
1.05
1.00
0.95
0.90
0.85
0.80
0.75
0.70
Donor-Grouped Split
Sample-Level Split (Leaked)
Raw HVG
PCA-64
cVAE
C Regional vulnerability & control
Metric Value
Progression ρ (95% CI)
AD vs Ctrl AUC
1.0
0.8
0.6
0.4
0.2
0.0
Entorhinal
Temporal
Frontal
Cerebellum
(Non-regional)